tarefa_cores

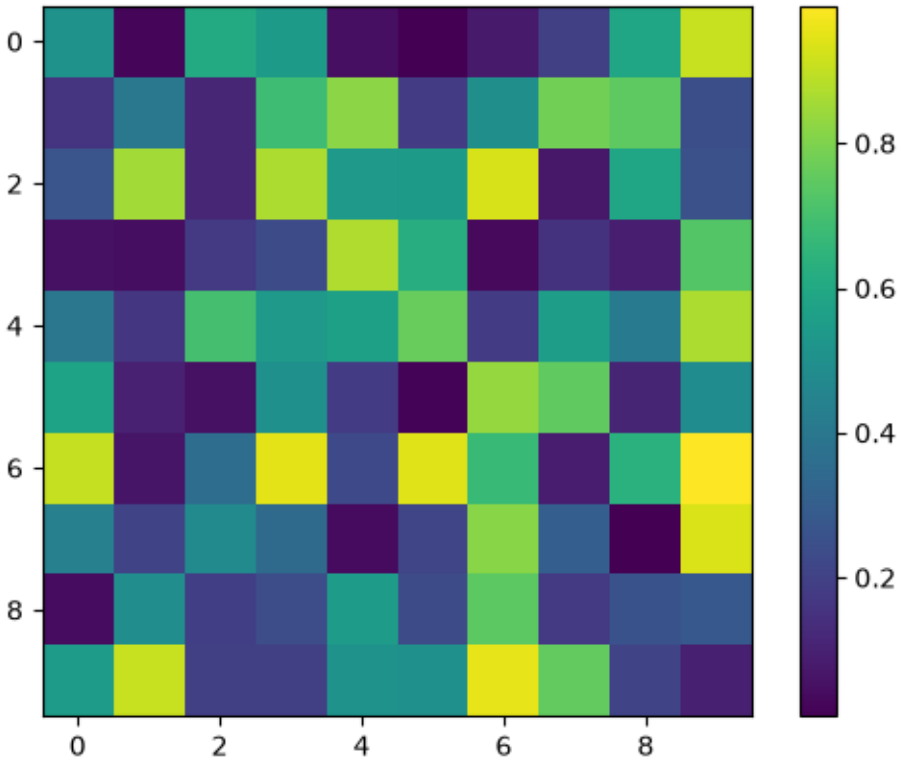
October 9, 2026

1. usando `viridis`

[24]:
```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng()
n = 10
A = rng.random((n, n))

plt.imshow(A, cmap='viridis')
plt.colorbar()
plt.show()
```



2. comparando dois colormaps

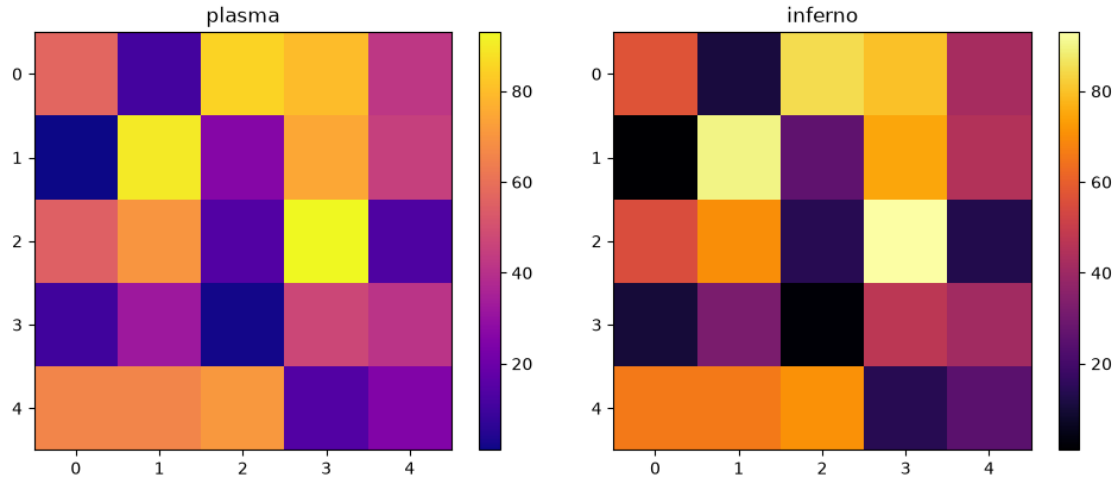
[34]:
```
n = 5
A = rng.integers(low=0, high=100, size=(n, n))

fig, axes = plt.subplots(1, 2, figsize=(10, 4))

im1 = axes[0].imshow(A, cmap='plasma')
axes[0].set_title('plasma')
fig.colorbar(im1, ax=axes[0])

im2 = axes[1].imshow(A, cmap='inferno')
axes[1].set_title('inferno')
fig.colorbar(im2, ax=axes[1])

plt.tight_layout()
plt.show()
```



3. invertendo um colormap

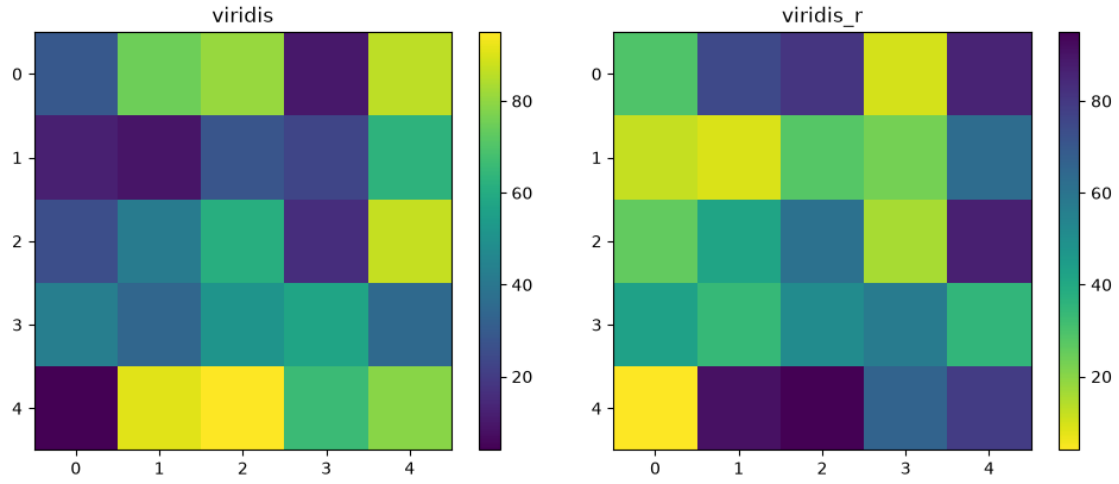
[37]:
```
A = rng.integers(low=0, high=100, size=(n, n))

fig, axes = plt.subplots(1, 2, figsize=(10, 4))

im1 = axes[0].imshow(A, cmap='viridis')
axes[0].set_title('viridis')
fig.colorbar(im1, ax=axes[0])

im2 = axes[1].imshow(A, cmap='viridis_r')
axes[1].set_title('viridis_r')
fig.colorbar(im2, ax=axes[1])
```

```
plt.tight_layout()
plt.show()
```

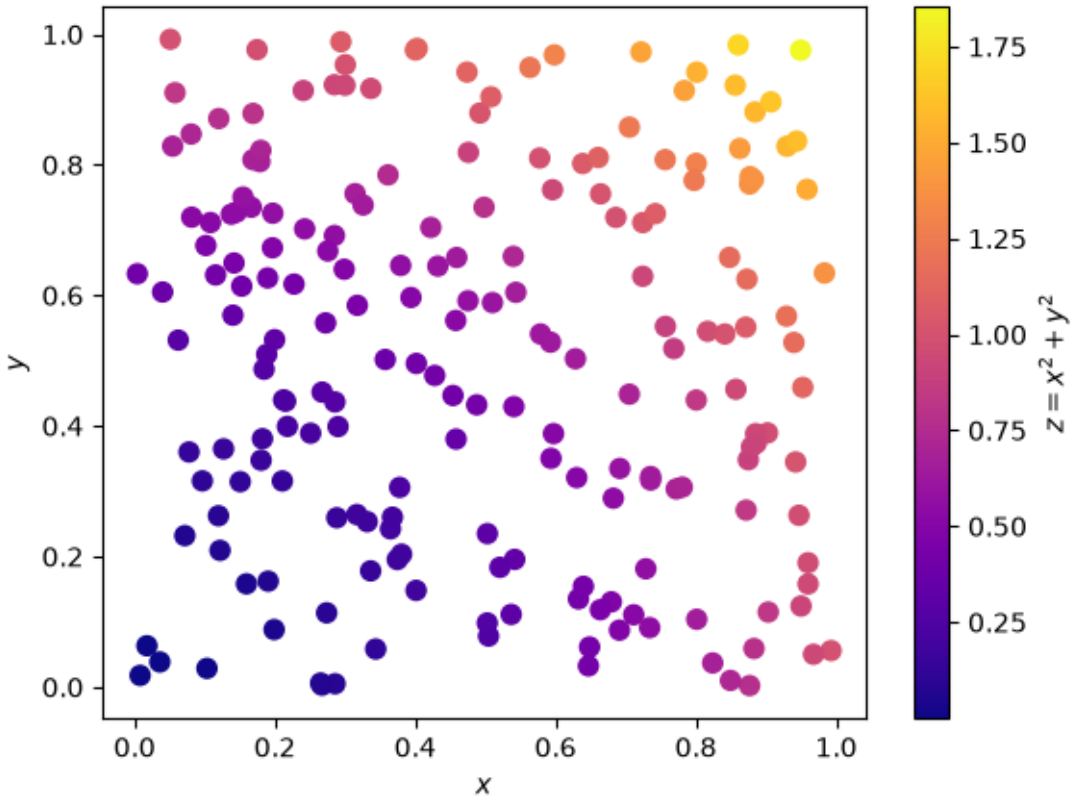


4. colormap em gráfico de dispersão

[42]:
```
x = rng.random(200)
y = rng.random(200)
z = x**2 + y**2

plt.scatter(
    x, y,
    c=z,
    cmap='plasma',
    s=50
)

plt.colorbar(label=r'$z=x^2+y^2$')
plt.xlabel(r'$x$')
plt.ylabel(r'$y$')
plt.show()
```



5. valores positivos e negativos

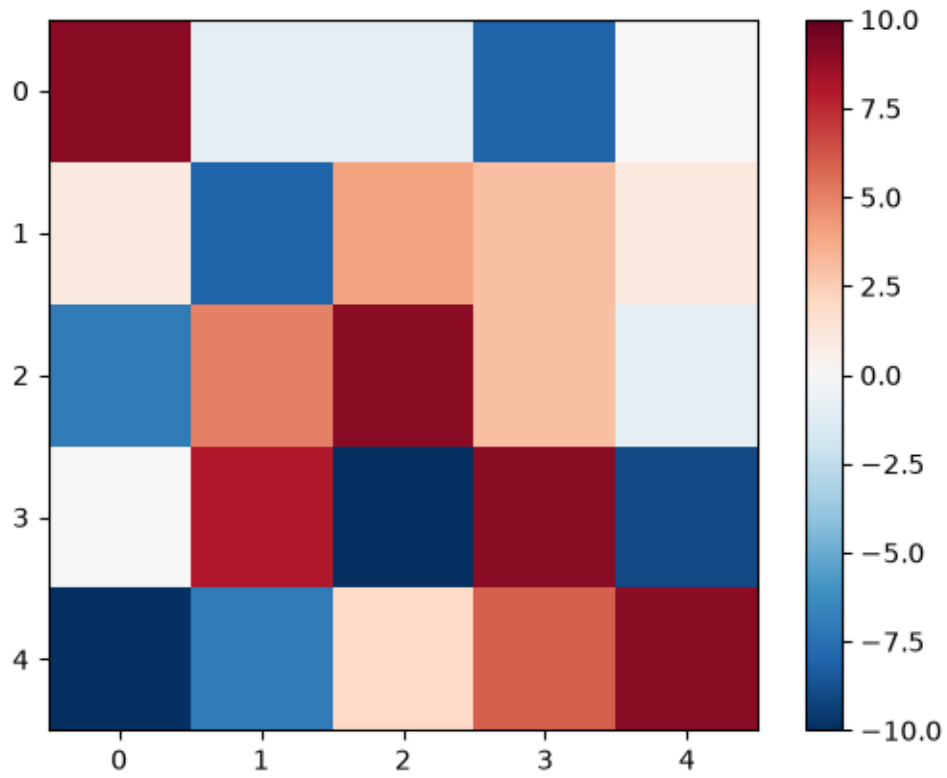
[43]:
```
from matplotlib.colors import TwoSlopeNorm

norm = TwoSlopeNorm(
    vcenter=0,
    vmin=-10,
    vmax=10
)

A = rng.integers(low=-10, high=10, size=(5, 5))

plt.imshow(A, cmap='RdBu_r', norm=norm)

plt.colorbar()
plt.show()
```



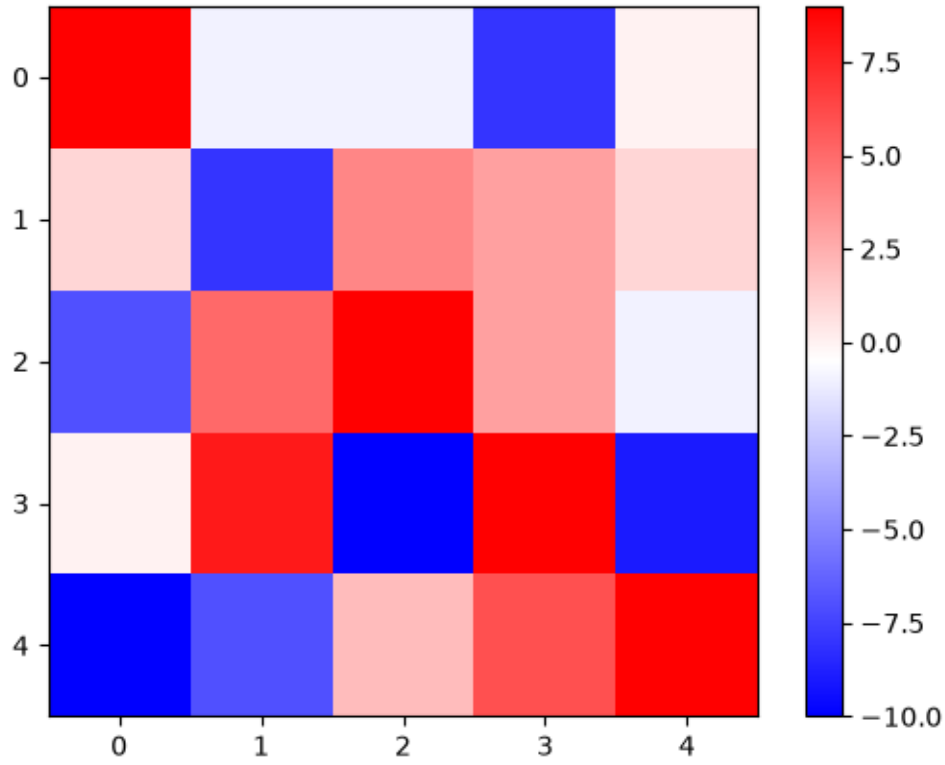
6. criando um colormap personalizado

[45]:
```
from matplotlib.colors import LinearSegmentedColormap

cores = ['blue', 'white', 'red']

meu_cmap = LinearSegmentedColormap.from_list('meu_mapa', cores)

plt.imshow(A, cmap=meu_cmap)
plt.colorbar()
plt.show()
```



7. colormap discreto

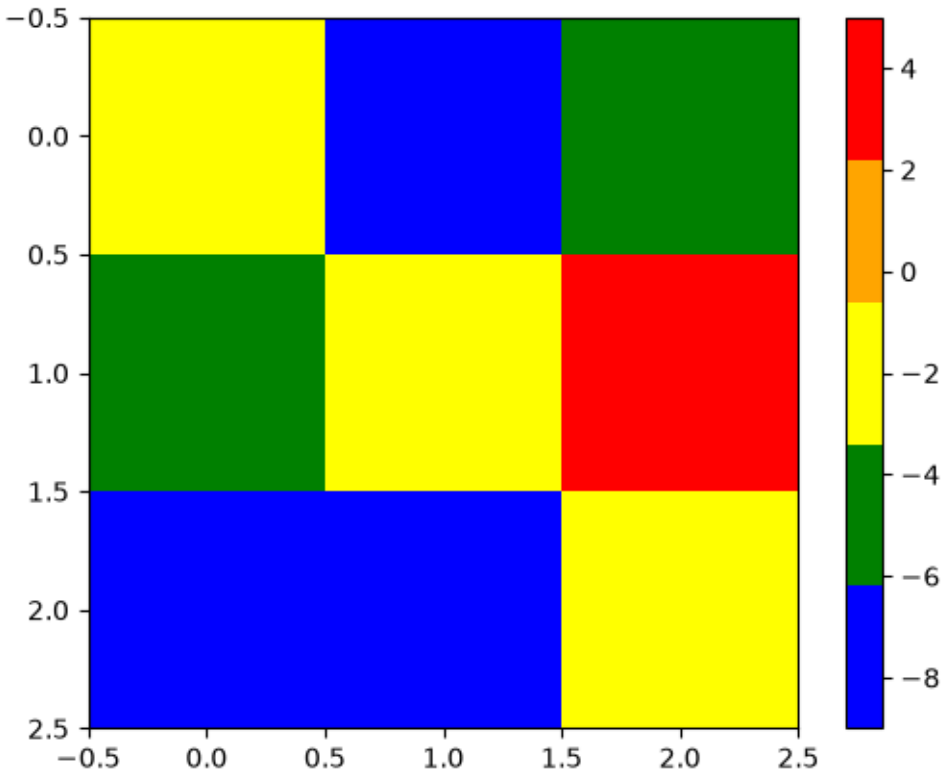
[98]:
```
from matplotlib.colors import ListedColormap

A = rng.integers(low=-10, high=10, size=(3, 3))

cores = [
    'blue',
    'green',
    'yellow',
    'orange',
    'red'
]

meu_cmap = ListedColormap(cores)

plt.imshow(A, cmap=meu_cmap)
plt.colorbar()
plt.show()
```



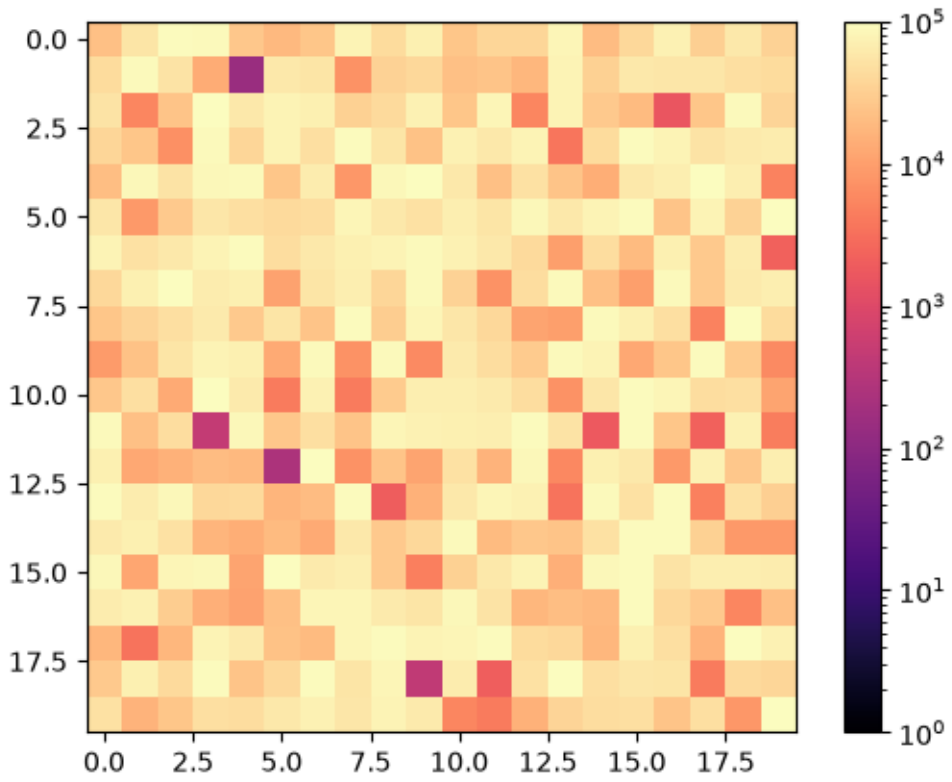
8. normalização logarítmica

[90]:
```
from matplotlib.colors import LogNorm

A = rng.integers(low=1, high=100000, size=(20, 20))

plt.imshow(
    A,
    cmap='magma',
    norm=LogNorm(vmin=1, vmax=100000)
)

plt.colorbar()
plt.show()
```



9. superfície 3D com colormap

[89]:
```
from mpl_toolkits.mplot3d import Axes3D

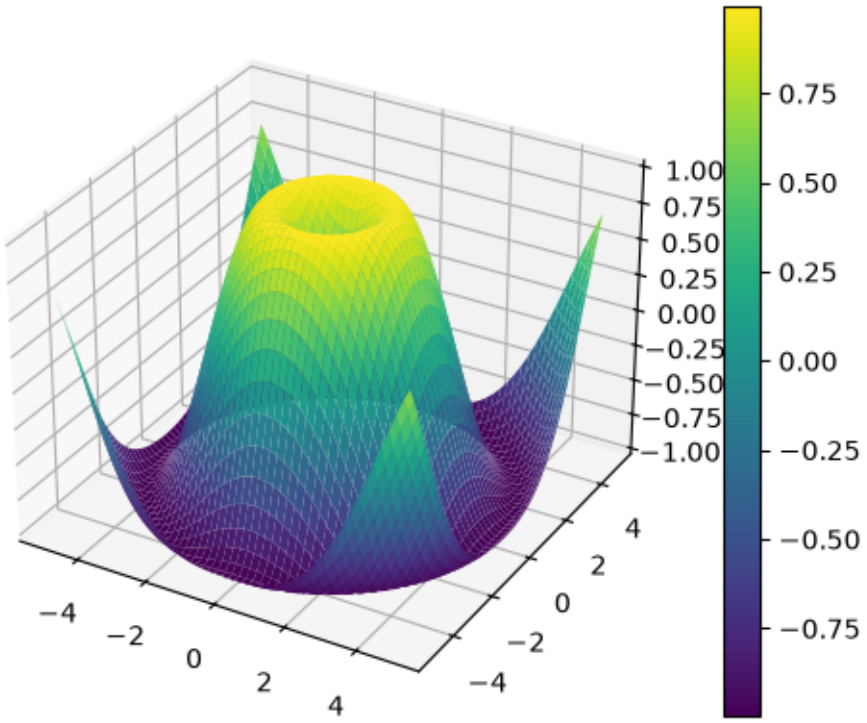
x = np.linspace(-5, 5, 200)
y = np.linspace(-5, 5, 200)

X, Y = np.meshgrid(x, y)
Z = np.sin(np.sqrt(X**2 + Y**2))

fig = plt.figure()
ax = fig.add_subplot(projection='3d')

surf = ax.plot_surface(
    X, Y, Z,
    cmap='viridis'
)

fig.colorbar(surf)
plt.show()
```



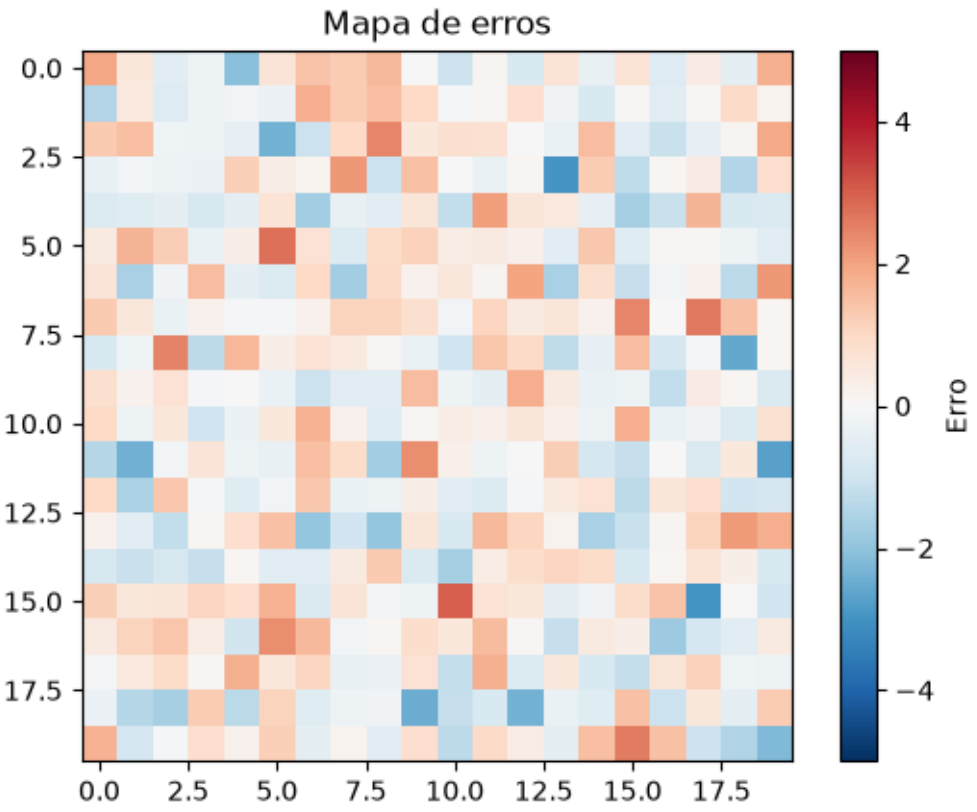
10. criando um mapa de erros

[96]:
```
norm = TwoSlopeNorm(
    vmin=-5,
    vcenter=0,
    vmax=5
)

erro = rng.normal(size=(20,20))

plt.imshow(
    erro,
    cmap='RdBu_r',
    norm=norm
)

plt.colorbar(label='Erro')
plt.title('Mapa de erros')
plt.show()
```


Mapa de erros
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
4
2
0
−2
−4
Erro